

FACULTY OF COMPUTING
UTM Johor Bahru

Group Project 3: Conceptual & Physical Design

SECV2113-HUMAN COMPUTER INTERACTION

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Section: 2

Contents

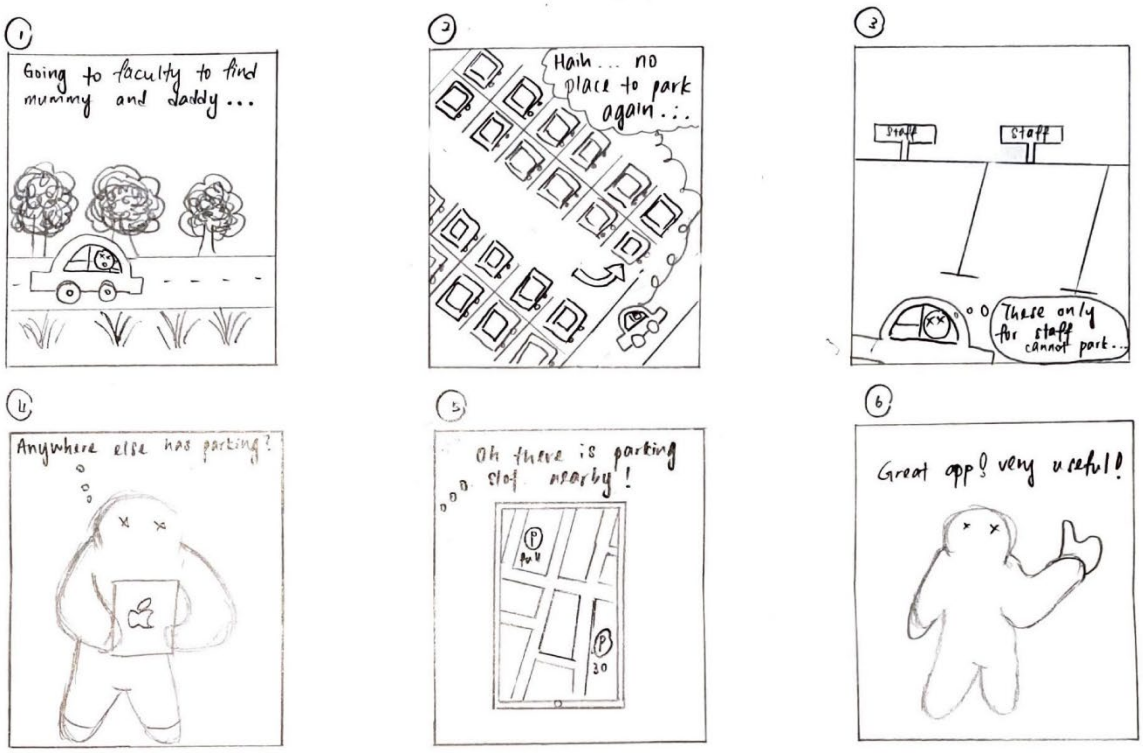
1.0 Storyboards 3
1.1 Task 1: Checking the bus schedule 3
1.2 Task 2: Checking the number of available parking spaces 4
1.3 Task 3: Making the report for the failure of the streetlight 5
2.0 Alternative Design 6
2.1 LIM SHI KAI 6
2.2 TAN CHUN MING 8
2.3 PHANG SENG SOON 10
2.4 DANIEL TENG WEI CHUNG 12
2.5 Scan of voted design layouts 14
3.0 Wireframes 15
3.1 Task 1: Checking the bus schedule 15
3.2 Task 2: Checking the number of available parking spaces 16
3.3 Task 3: Making the report for the failure of the streetlight 17
3.4 Justification of Design 18
4.0 Description and justification of metaphors 19

1.0 Storyboards

1.1 Task 1: Checking the bus schedule



1.2 Task 2: Checking the number of available parking spaces



Scene 1

The user is driving and heading to the faculty to find his parents to have lunch together.

Scene 2

The user cannot find a parking slot after finding 20 minutes.

Scene 3

Some of the parking slots are empty, but the user cannot park there due to that slot is reserved for the staff only.

Scene 4

The user opens the application to find whether there is any parking space nearby instead of the parking space he is currently at.

Scene 5

The map inside the application shows there is another parking space that has a parking slot available.

Scene 6

The user feels that the application is very useful and user-friendly.

1.3 Task 3: Making the report for the failure of the streetlight



Scene 1
The user is driving at night time.

Scene 2
The user drives into an area where the street lights are not functioning.

Scene 3
Suddenly an animal walk across the road and the user almost hit the animal.

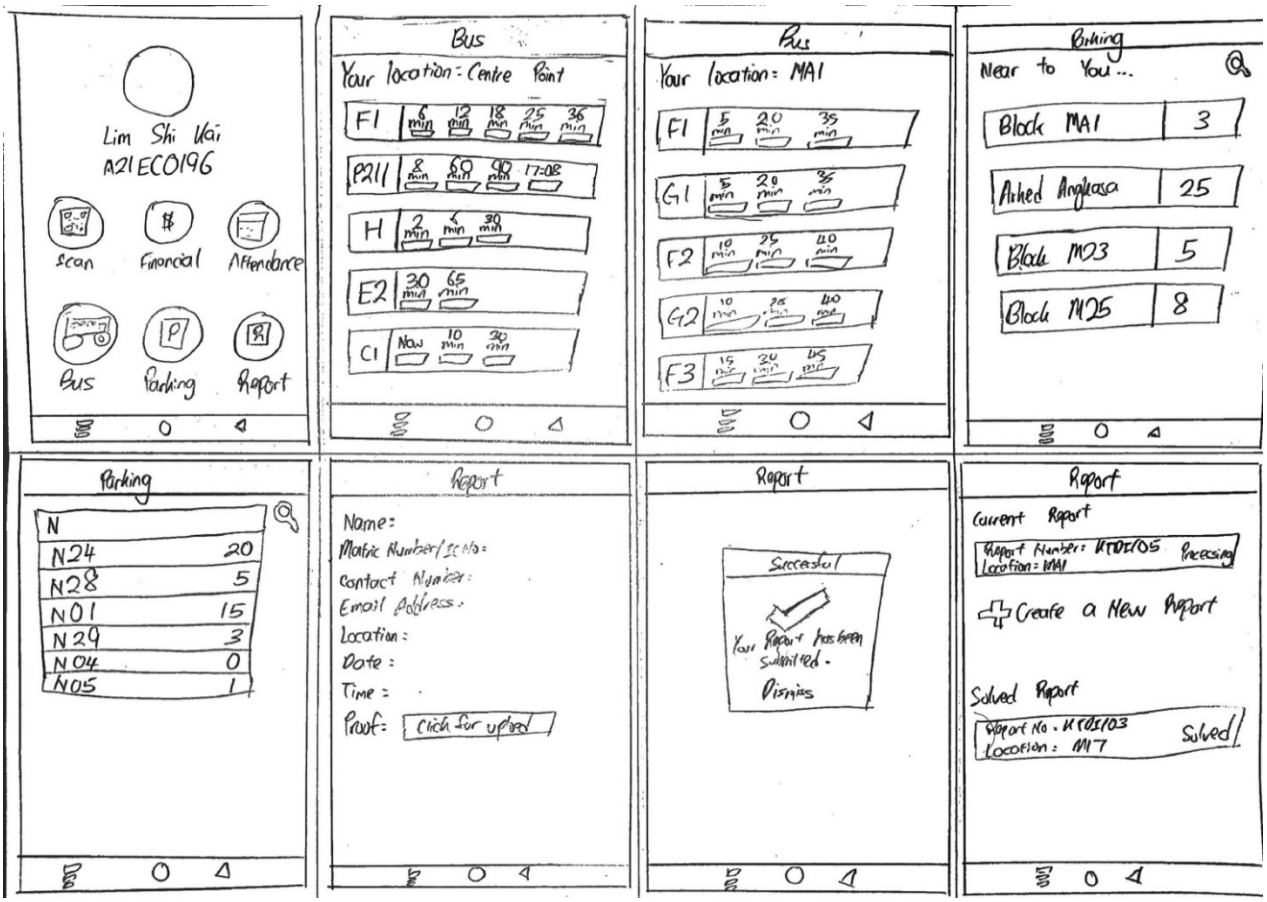
Scene 4
The user opens the application to do a report about this issue.

Scene 5
Inside the report, the user needs to fill in some information needed

Scene 6
The technician received instructions and go to repair the street lights.

2.0 Alternative Design

2.1 LIM SHI KAI



Screen 1

This page is the main page of our UTM Rocket app. The user can choose the function they want to use on this page like scan, financial, attendance, bus, parking and also report. Meanwhile, the profile picture and short information of the user will be shown here.

Screen 2

This page is one of the pages for the bus page. When the user opens the app, then the app will auto-locate the place where the user is. The system will show the estimated arrival time for each bus and the status of the spaces in the bus. So, this page is shown that the user is waiting for the bus at the centre point station, which is the hub for all the bus routes. Besides that, the P211 bus, which is the bus to fetch the students in or out from UTM, also will know the ETAs in our app.

Screen 3

This is the page that shows the bus around students' resident college. The bus page will be updating the page from one point to one point, which can give a better experience for our users.

Screen 4

This page is showing that the user is driving his car and plan to park in its college block area. So, he opens the app, the app will be in the parking lots near him, like some residential block or office area. The sequence for the parking lots that are shown will be arranged from nearest to farthest from the location point.

Screen 5

Besides the auto location, the user also can search the parking spaces that are still available at their destination. Once a keyword is input in the search bar, the area that started from the specific keyword will be shown, to let the user know the parking spaces in his input.

Screen 6

This page is the page where the user clicks on the report icon on the main page of the UTM Rocket app. This page will be shown the user clicks the "Create a new report", and then the user can file the report when the user is noticing the failure streetlight around the UTM campus. The user is required to fill up their personal details, date, time and the location of the failed streetlight. The user was also urged to upload the photo as proof of report.

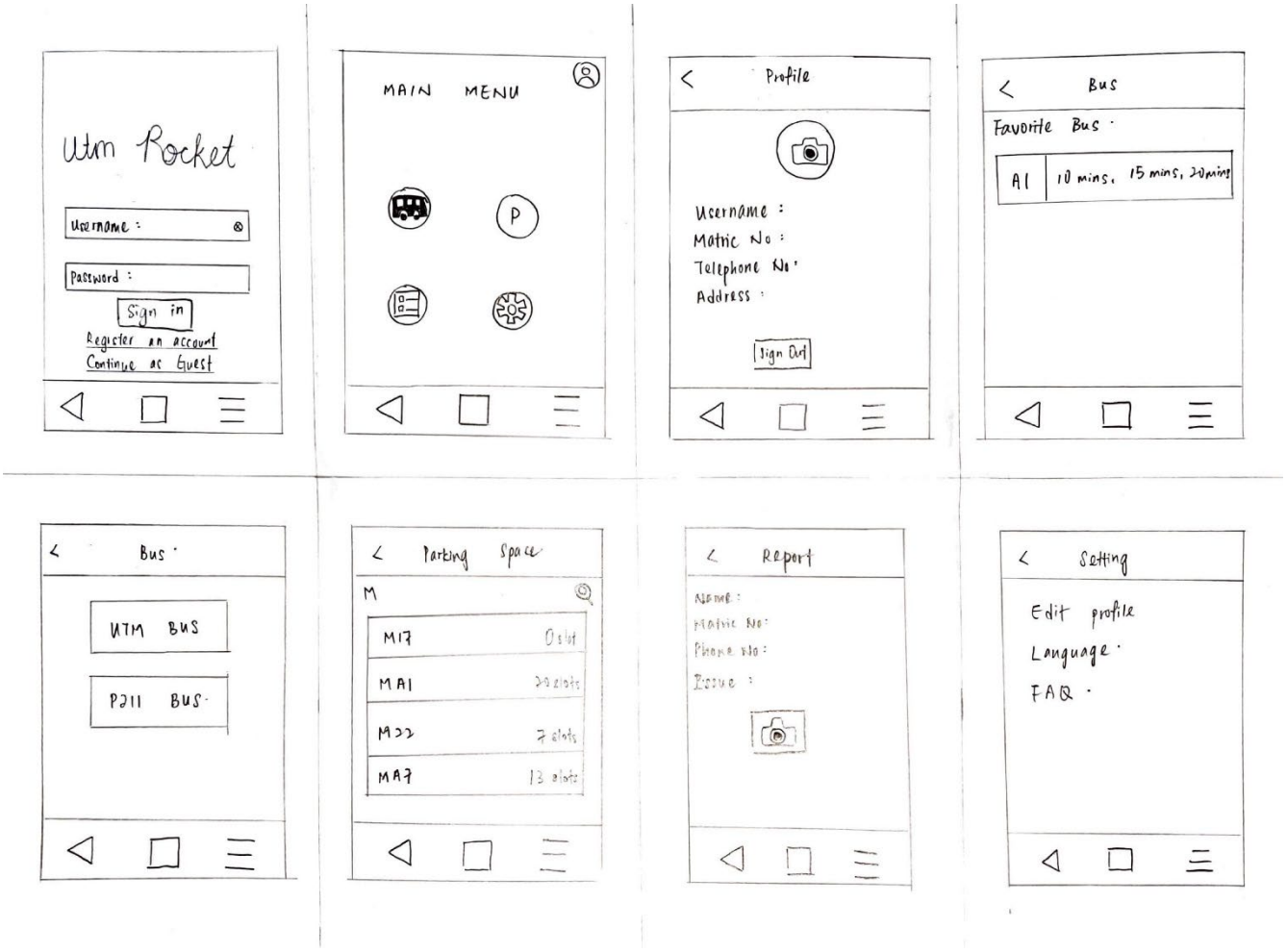
Screen 7

This page shows that the report is submitted successfully, and the report number will be given to the reporter for future progress updating purposes.

Screen 8

This page is the main page for the report function. The report that was made by the user will be shown here with the current status like "Submitted", "In Progress", or "Closed". Then, the user also allows making a new report.

2.2 TAN CHUN MING



Screen 1

This page shows the default start-up page of the application which is the sign-in page. On this page, there are three main parts, which are the sign-in, sign up and continue as a guest, where sign-in is for the existing user to log in, sign-up is for the new user registration and then continue as the guest is used for the user who just wants to use the application temporarily.

Screen 2

This page is the main page or the home page of the application which contains four main functions which will bring the user to the respective page such as bus, parking space, report and setting. There is also a profile icon at the top right corner which users can click to see their profile.

Screen 3

This page is the profile page of the users once they registered and log into their accounts. There will be showing the details of the user such as the profile picture of the user, username, matric

number, phone number and addresses. There is also a sign-out button which allows users to log out of their accounts.

Screen 4

This page is the page where users can find the bus schedule. The main point of this page is that users can highlight a specific bus schedule so that it will reduce the time cost to find the bus schedule which is often searched by the user and users do not have to waste their time to find the bus schedule among all of the bus schedules.

Screen 5

This is the page where the application will display the types of bus schedules that the user would like to search for. In our case, there will be two main categories of the bus schedule: the UTM bus and the P211 bus provided by the Johor government. By clicking the respective category, the application will bring the user to the page that consists of the bus schedule respectively.

Screen 6

This page is the page for finding parking space. The application will be showing the list of the parking space searched by the keyword and the table will show the parking slot available in the parking space respectively. There is also a clear button for the user to clear the keyword searched and back to the page which consists of all of the parking spaces nearby.

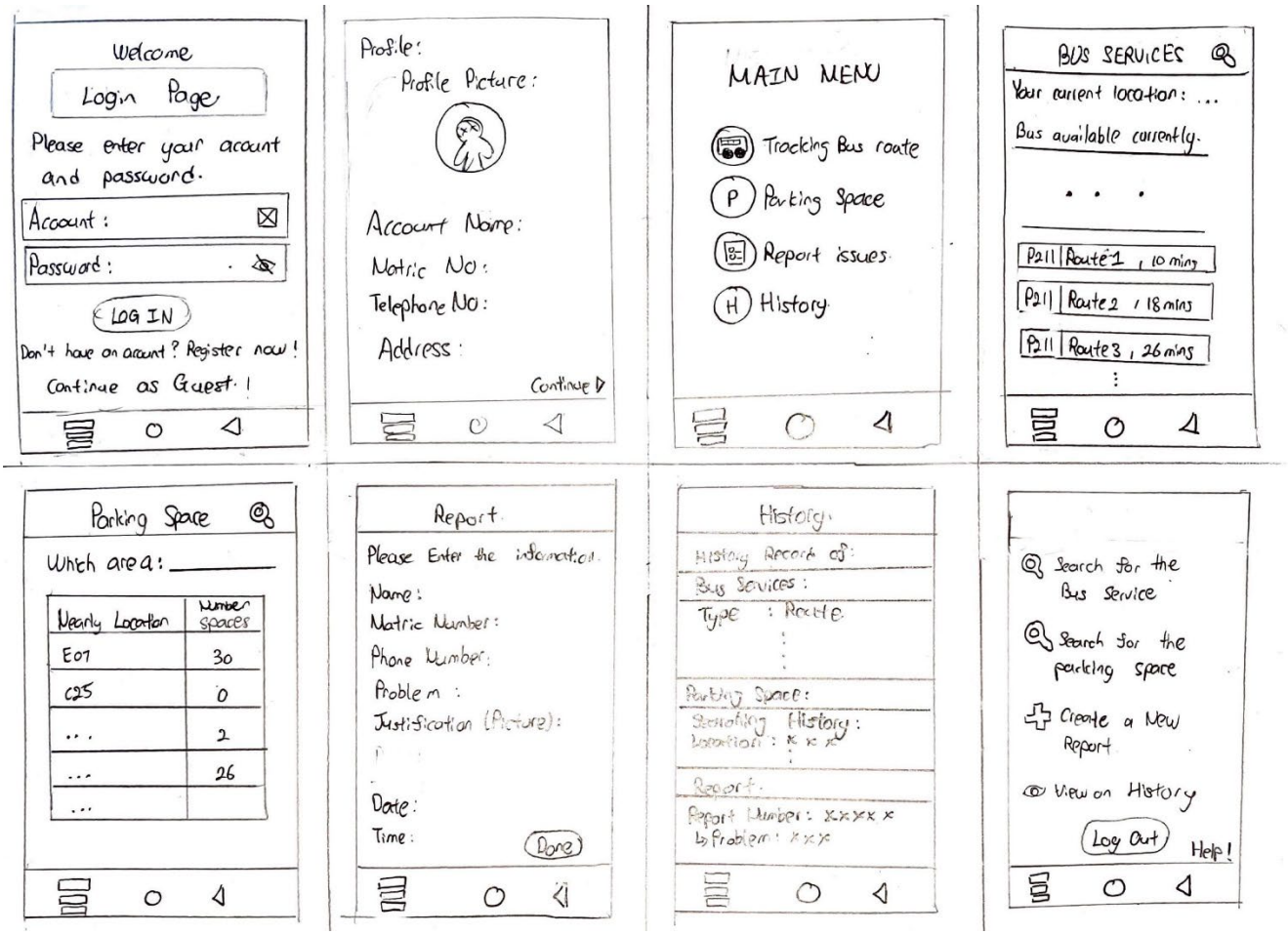
Screen 7

This is the page the user wants to make a report about an issue. On this page, there will have some information that requires the user to insert such as the name, matric number, phone number, and issue that the user wants to report. There is also a button where the user can take a photo as proof of the issue.

Screen 8

This page is the setting page of the application. In this page, it consists of three main functions which are the edit profile where the application allows the user to edit their profile, the language button which will be more user-friendly considering that there might be foreign visitors who do not read both English and Malay well and the frequently asked question (FAQ) which the function consist of some commonly asked questions and the respective answers.

2.3 PHANG SENG SOON



Screen 1

This login page allows the user to access other applications by entering their username and password for authentication purposes. When entering the username in the “Account” row, users are allowed to click the cross sign to delete the username quickly in case a typo has happened. For passwords, the user can click to view all the passwords for checking purposes. If the users don’t have an account, they can click on “Register now” to create an account for this application or by selecting a guest to continue using it. Once the username and the password have been keyed in, users are required to click on the "login" button to enter the application.

Screen 2

This page is used to show the profile of the user by displaying the account name, matric number, telephone number, and address as well as the profile picture that has been set by the user every time the user enters the application. By clicking the "Continue" button, the user will be directed to the main menu screen.

Screen 3

The screen of the main menu, it was showing some functions provided in this application. Users can click on the icon or the text to be directed to the services function. The first icon is the bus. The second icon is for the parking space. The third icon is for the user to make a report on the issues that happened. For the fourth icon is the history button.

Screen 4

If the user clicks on the bus icon or the text beside the icon, the user will enter the bus services page. This page was used to track the bus route and get the estimated time of the certain bus. Besides that, this application will try to show the current location in which the user is located to make sure the location is correct and not skewed by any uncontrollable factors.

Screen 5

For the parking space page, the user is allowed to enter the location that they want to go, and then the application will display all the nearly available parking spaces around the location to provide alternative choices to the user. Not only that, but it will also show the number of parking spaces that are still available for that place.

Screen 6

For the report page, when the user wants to report any issues, the user will be asked for entering the name, matric number, phone number, problem, and justification which is a picture as proof to show that the reality of this report, and the date and the time when realizing this problem, then user need to click done to submit the report successfully.

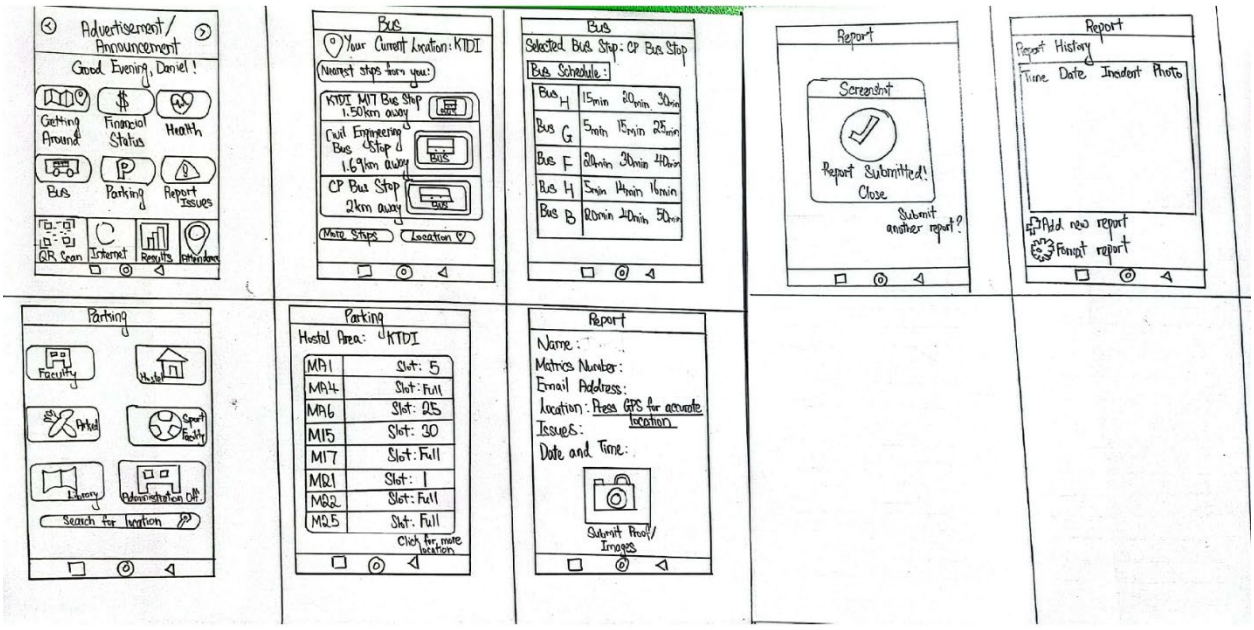
Screen 7

For the history page, the user can view the record based on the functions provided respectively. It will show the “bus services”, “parking space”, and the “report” tab. Once the tab has been clicked on, then it will display the information of the record that the user searched and clicked previously.

Screen 8

At the end of this page, it will display all the services again for the user to make the process convenient because the user does not need to click back and back to reach the main menu page. The “Help” button at the bottom of this page is used for the user to get the information regarding how to set up according to their preferences and understand the details of the functions and it allows the user to get the information regarding this application in case they are not familiar with this application on how to use it. The “Log out” button at the bottom is for a user to quit the account and exit the application.

2.4 DANIEL TENG WEI CHUNG



Screen 1

It shows the proposed front page or main page of the mobile application. On this front page, there are actually a series of functions and buttons that have been displayed in an orderly manner, and there are 6 main functions that we already included in the main page, to further ease the usability of the users, where they can only just access that specific area only in one click, without having to further navigate to other panels or selections. Besides that, 4 main functions used by the students, which are QR Scan, Internet Usage, Results and also attendance, are listed in the menu segments so that they can also directly access all these functions in one click.

Screen 2

It shows the number of bus stops available at the user's nearby location, and also at the same time showcases the duration of that bus stop away from the user. It also has a selection which enables users to check for more bus stops allocated and dispersed around the campus, and at the same time able to track the bus schedule of that specific bus stop. Besides that, students can also track the location of the bus stops by pressing on the location selection, and they will be redirected to the map that displays the exact location of bus stops.

Screen 3

On-screen 3, clearly shows the timetable or schedule of bus services at that specific bus stop. It also clearly shows the estimated time of arrival of that particular bus shuttle, and also at the same time will show the exact amount of bus shuttles that are still in service that will stop at that specific bus stop.

Screen 4

In this parking segment, users can navigate freely inside the section, and at the same time, they can also choose where they are looking or searching for parking. For instance, faculty, hostel, arked, sports facility, library or administrative office are the locations where the users can track the parking, and at the same time, they can also instant search for the location, once they clearly know the specific location that they are heading towards.

Screen 5

For instance, if the students selected a hostel selection, and the hostel area is detected around the KTDI area, there will be a list of the hostels that are sparsely located around the KTDI area. The hostel number or block number will also be arranged in an orderly manner, and each hostel will also be displayed the exact amount of parking left in that specific location or hostel block so that the users can be alerted on a time basis regarding the availability of the parking slots in that specific hostel block.

Screen 6

This report-maker segment allows users to lodge a report regarding the issues or incidents happening surrounding them that required further assistance. For instance, the system will require the name, matrix number and also email address of the students or users and will also allow users to use the GPS location function to directly mark the location, and will also allow users to specify the issues together with date and time. The users will also need to submit an image as proof.

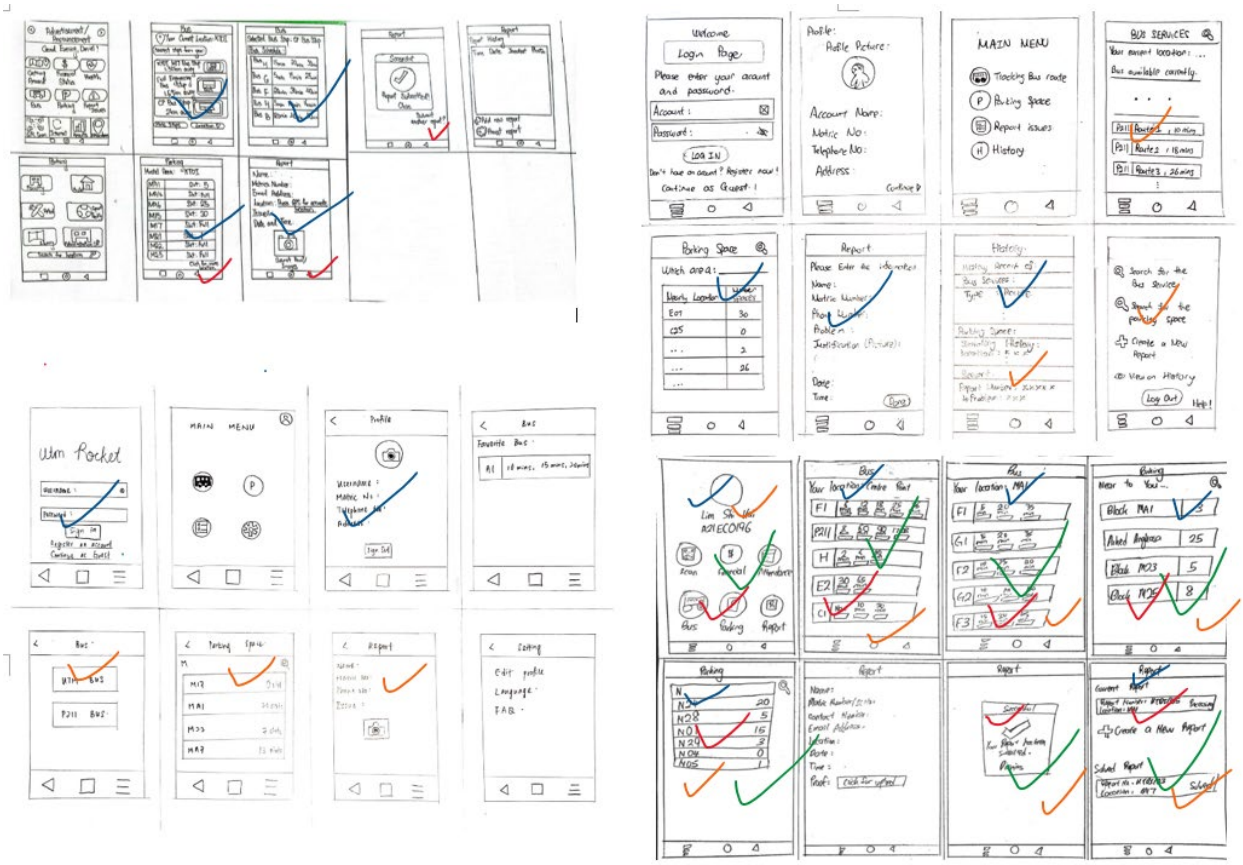
Screen 7

Once the user successfully submitted the report with all the accurate and correct details, the system will pop up a successful notification that indicates to the users that they had successfully lodged the report. They will also be able to lodge another report to further report other incidents or issues.

Screen 8

The reporting system will also have a history of all the reports made by the users, and also have functions that enable users to lodge additional reports, and at the same time, users can format their reports based on the settings functions.

2.5 Scan of voted design layouts



1. LIM SHI KAI – vote for everyone

Reason: He wants the good things of everyone's design be combined together and become a better interface.

2. TAN CHUN MING – vote for LIM SHI KAI

Reasons: His design is very detailed and oriented to the problems we have discovered and discussed I in part 2. His design has solved the problems we discovered very detail.

3. PHANG SENG SOON – vote for LIM SHI KAI + DANIEL TENG WEI CHUNG

Reason: Their design is very detailed design and shows the very informative details to the users.

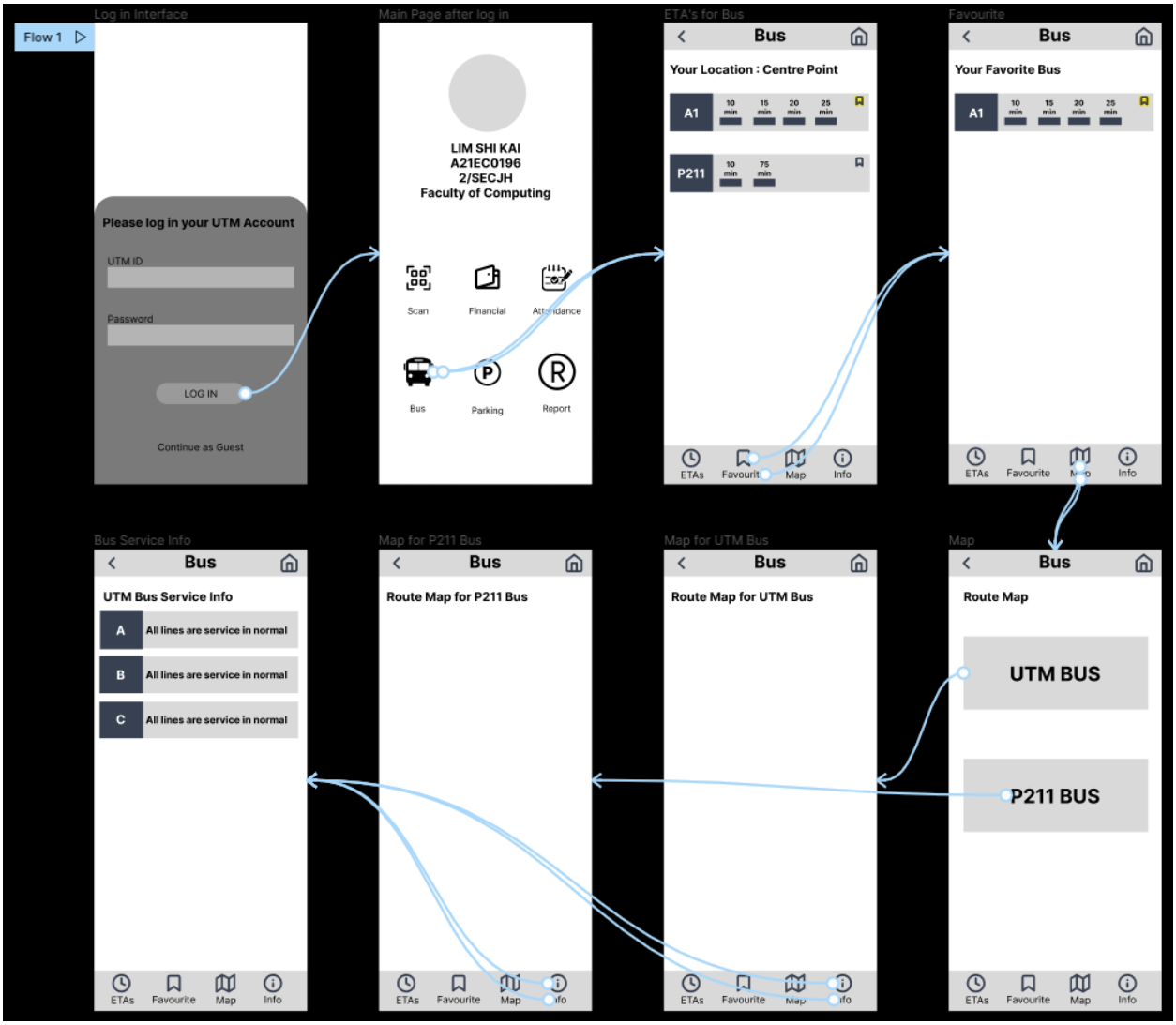
4. DANIEL TENG WEI CHUNG – voted for everyone.

Reason: He wants the good things of everyone's design be combined together and become a better interface.

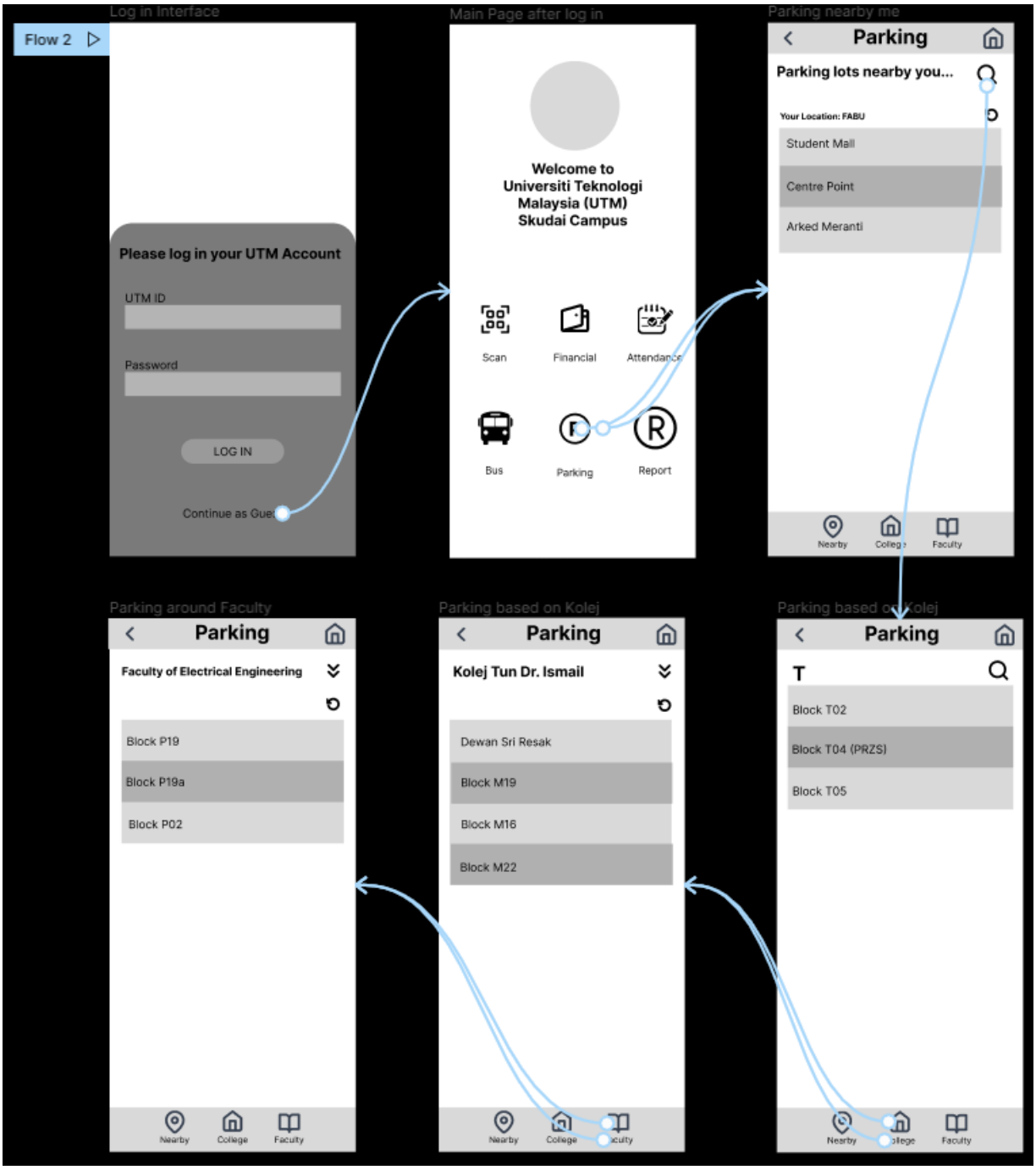
Result: We all decided to use LIM SHI KAI's design as a basic. Then, we refine it by combining all the good part of everyone's design.

3.0 Wireframes

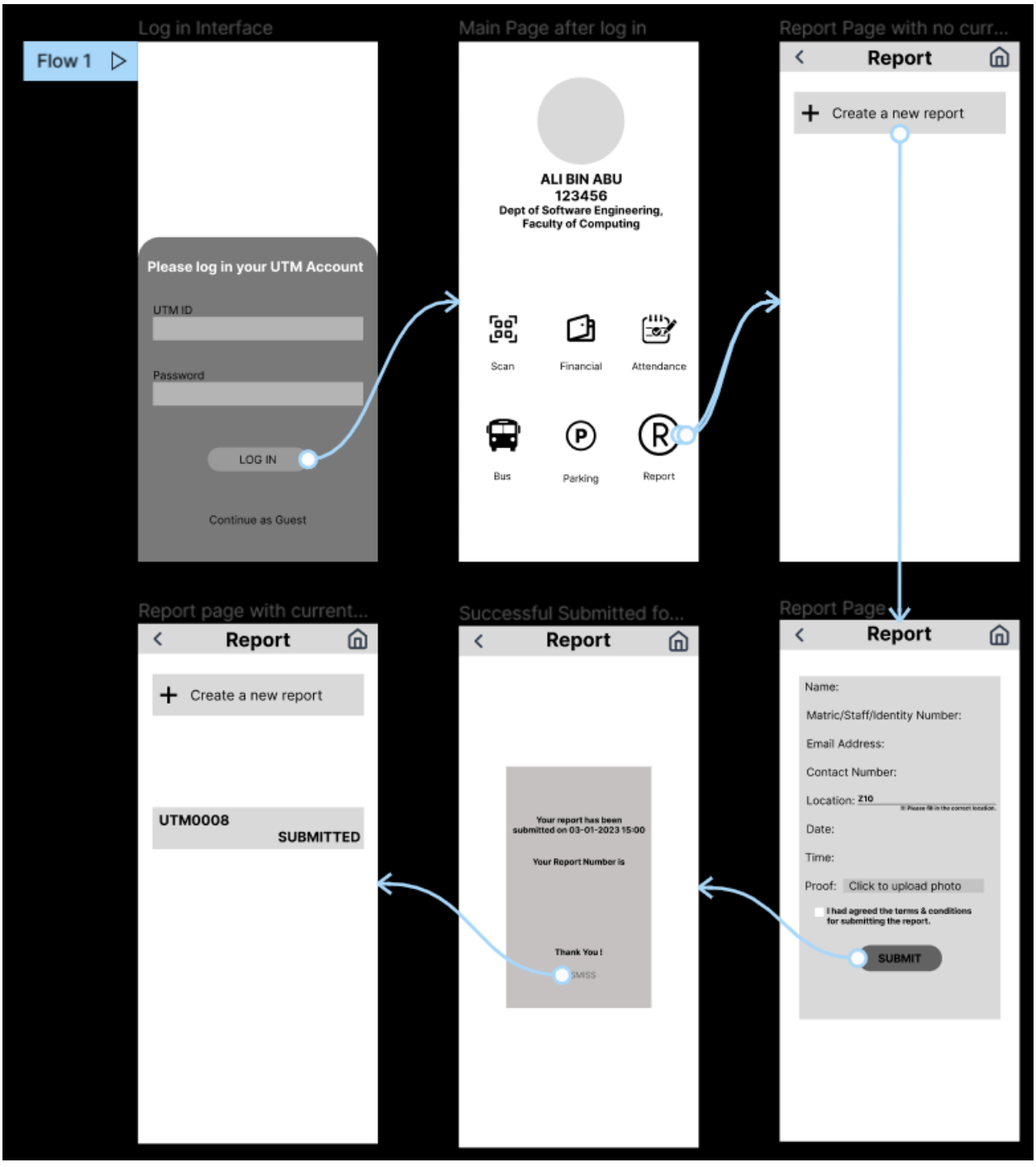
3.1 Task 1: Checking the bus schedule



3.2 Task 2: Checking the number of available parking spaces



3.3 Task 3: Making the report for the failure of the streetlight



3.4 Justification of Design

In our system, we implemented four Gestalt principles, throughout the whole design, which are Similarity, Proximity, Continuation and Figure & Ground. For here, we will only pick some interfaces to show the Gestalt principles.

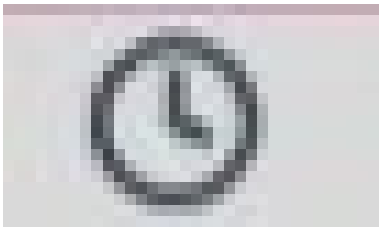
For similarity, we can see it on our main page after the login page. For every app, we use the same shape and design for the app icon. Then, in the design of apps, we use icon design that most the people recognize and prefer, like the parking icon that can be found in our daily lives for the parking button. We can also see Proximity in the main interface. We group the function icons in two rows, and then we put the information which is the student or staff name, matric or staff number at the top centre of the main page. Besides, we can also see the Continuation on the main page of our app. We use the original function that has in the UTM Smart app which are scan, attendance and financial status, users still can use the function in UTM Smart in our app, no need to install both apps at the same time. Finally, Figure and Ground are shown in our app too.

Besides that, our app has shown some principles which are from Shneiderman's 8 Golden Principles. The principles are to Strive for Consistency and reduce short-term memory load, in which we are utilising the familiar icons and colours in our app. As an example, the bus and parking icon is used in our app, and the users can know the icon refers to what function that provided in our app. Next, we also enable frequent users to use shortcuts in our app, which the users can just click the home icon to return to the homepage in our app and process the task that wishes to.

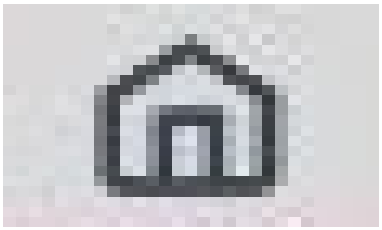
In other that, the UTM Rocket app also offers informative feedback to the users. When the users are filling the report column with inappropriate information, our app will be shown the line of the filled part in red colour to notify the user that the information that is keyed in might be wrong. The design dialogue to yield closure principle also contains in our app, the user will be received the report number with "Thank You" after they submit the report in our app.

Lastly, the users permit easy reversal of actions, in which the users can click the home icon to return to the homepage, even if they are having multiple steps in the app, then just a click can return to the homepage for logging out or processing other functions. In the UTM Rocket, the users support the internal locus of control, which means the users are given full control of the apps without any restrictions by the designer or developer.

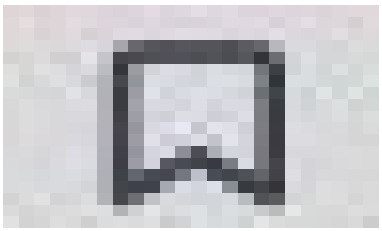
4.0 Description and justification of metaphors



Clock in Menu Bar
Functions: To showcase the timetable of the bus schedule according to the bus stops that have been selected



Home
Functions: To redirect the users back to the homepage of the mobile application



Bookmark
Functions: Enable users to bookmark and save their favourite bus stops, and make the users more convenient in searching for their all-time favourite bus stops



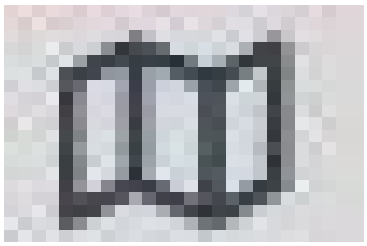
Bus
Functions: Direct and channel users to the bus shuttle schedule segments

the 

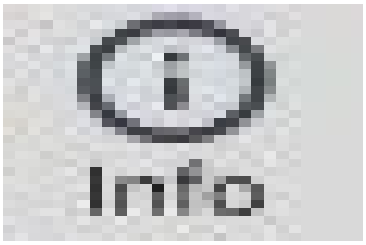
Parking
Functions: Direct and channel users to a smart parking system where users can freely navigate the number of parking spaces surrounding them



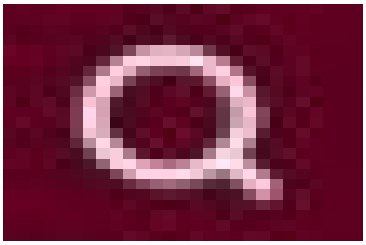
Report Sign
Functions: To direct users to the report section that allows them to lodge issues or incidents via the report system



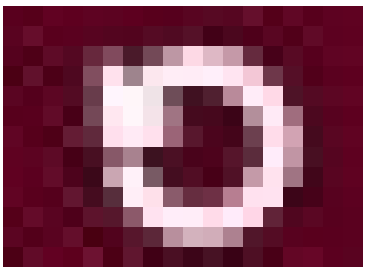
Map Sign
Functions: To let users check on the bus route map regarding bus P211 and also the shuttle bus in UTM



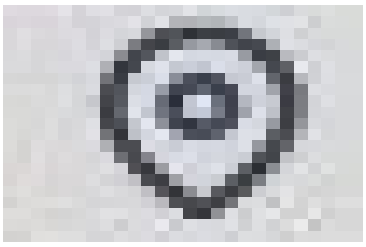
Info Signage
Functions: To enable users to click on this section and get first-hand information regarding all the bus services and their respective routes in a more detailed manner



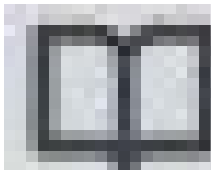
Search Icon
Functions: To enable users to click on the icon and directly search for the specific location that they wished to navigate the number of parking slots available in that specific area



Reload Icon
Functions: To refresh the number of parking slots available and at the same time able to get first-hand updates and information regarding the parking slots available



Location Icon
Function: To allow users to search for the nearby location available for parking with their GPS being activated



Book Icon
Function: To allow users to search for respective faculties parking slots, as we metaphorized book as faculty because faculty is the place where students studied